



COMPUTER
ENGINEERING

QUEZON CITY

COURSE CODE/TITLE: CPE212 | Automating
Management Server

SECTION: CPE31S2

DATE PERFORMED: 08/13/2026

DATE SUBMITTED: 08/13/2026

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ACTIVITY NO. **PRELIM SKILLS EXAM**

1. PROCEDURE OUTPUT

What procedures did you perform?

PRELIM

```
iquin@workstation: ~/ansible-lab
the implicit localhost does not match 'all'
hosts (0):
iquin@workstation:~$ sudo nano /etc/ansible/hosts
iquin@workstation:~$ sudo nano /etc/ansible/hosts
iquin@workstation:~$ mkdir ~/ansible-lab
iquin@workstation:~$ cd ~/ansible-lab
iquin@workstation:~/ansible-lab$ nano inventory.yaml
iquin@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
```

```
iquin@workstation:~/ansible-lab$ nano inventory.yaml
iquin@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
  "_meta": {
    "hostvars": {
      "centos1": {
        "ansible_host": "192.168.56.125"
      },
      "ubuntu1": {
        "ansible_host": "192.168.56.123"
      },
      "ubuntu2": {
        "ansible_host": "192.168.56.124"
      }
    }
  },
  "all": {
    "children": [
      "ungrouped",
      "ubuntu",
      "centos"
    ]
  },
  "centos": {
```

```
iquin@workstation: ~/ansible-lab
"ansible_host": "192.168.56.124"
    }
  },
  "all": {
    "children": [
      "ungrouped",
      "ubuntu",
      "centos"
    ]
  },
  "centos": {
    "hosts": [
      "centos1"
    ]
  },
  "ubuntu": {
    "hosts": [
      "ubuntu1",
      "ubuntu2"
    ]
  }
}

iquin@workstation:~/ansible-lab$
```

Lists of hosts

```
ping : pong
}
iquin@workstation:~/ansible-lab$ ansible all --list-hosts
hosts (3):
  192.168.56.123
  192.168.56.124
  192.168.56.125
iquin@workstation:~/ansible-lab$
```

5. Pinging managed nodes

```
iquin@workstation: ~/ansible-lab
iquin@workstation:~/ansible-lab$ nano inventory.yaml
iquin@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
ubuntu2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
centos1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
iquin@workstation:~/ansible-lab$
```

PART C

1. Modifying the inventory file

```

GNU nano 7.2                                inventory.yaml
all:
  children:

    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.123
        ubuntu2:
          ansible_host: 192.168.56.124

    centos:
      hosts:
        centos1:
          ansible_host: 192.168.56.125
          ansible_user: melquisedec

    linuxservers:
      children:
        ubuntu:
        centos:

[ Read 20 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Locatio
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To L

```

2. Pinging the metagroup

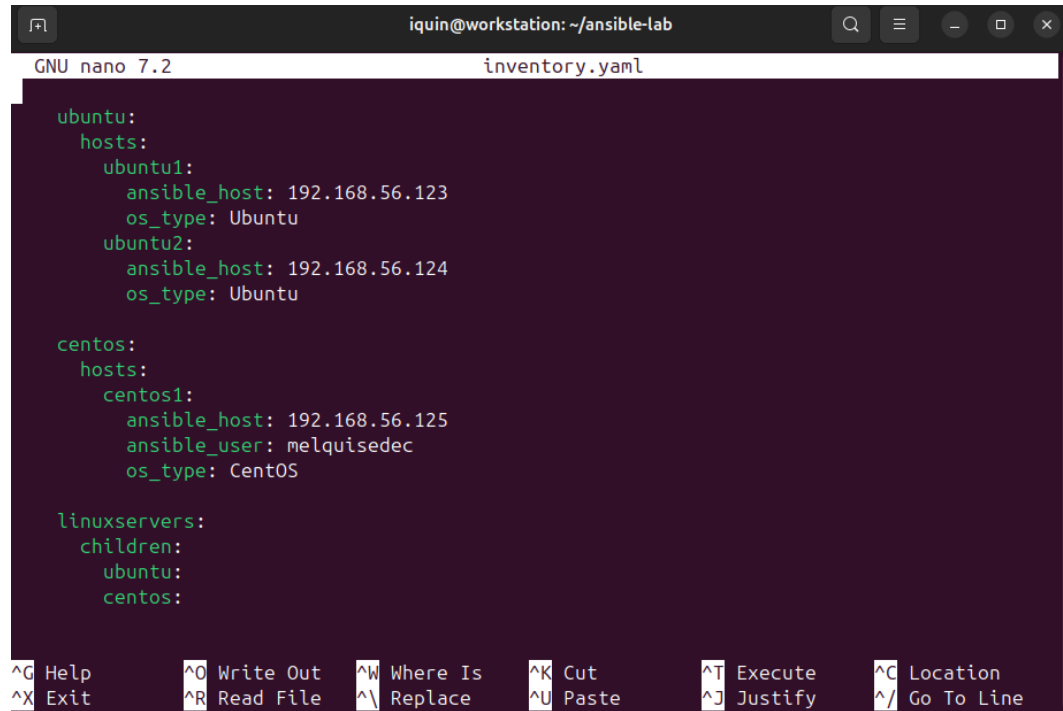
```
iquin@workstation: ~/ansible-lab
iquin@workstation:~/ansible-lab$ nano inventory.yaml
iquin@workstation:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yaml
ubuntu2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
centos1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
iquin@workstation:~/ansible-lab$
```

3. Displaying the hosts

```
iquin@workstation:~/ansible-lab$ nano inventory.yaml
iquin@workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.
yaml
hosts (3):
  ubuntu1
  ubuntu2
  centos1
iquin@workstation:~/ansible-lab$
```

PART D

1. Adding host variables



```
iquin@workstation: ~/ansible-lab
GNU nano 7.2 inventory.yaml

ubuntu:
  hosts:
    ubuntu1:
      ansible_host: 192.168.56.123
      os_type: Ubuntu
    ubuntu2:
      ansible_host: 192.168.56.124
      os_type: Ubuntu

centos:
  hosts:
    centos1:
      ansible_host: 192.168.56.125
      ansible_user: melquisedec
      os_type: CentOS

linuxservers:
  children:
    ubuntu:
    centos:
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

2. Adding group variables

```

iquin@workstation: ~/ansible-lab
iquin@workstation:~/ansible-lab$ nano inventory.yaml
iquin@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]
ok: [centos1]

TASK [Show Operating System] *****
ok: [server1] => {
  "msg": "Host server1 is running Ubuntu"
}
ok: [server2] => {
  "msg": "Host server2 is running Ubuntu"
}
ok: [centos1] => {
  "msg": "Host centos1 is running CentOS"
}

PLAY RECAP *****
centos1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
iquin@workstation:~/ansible-lab$

```

PART E

1. Creating playbook

```

iquin@workstation: ~/ansible-lab
GNU nano 7.2      playbook.yaml *
---
- name: First Ansible Playbook
  hosts: linuxservers

  tasks:
    - name: Ping all managed nodes
      ping:
    - name: Display Hello World
      debug:
        msg: "Hello World from {{inventory_hostname}}"
    - name: Display Operating System
      debug:
        msg: "Operating System: {{os_type}}"

```

2. Execute the playbook

```
iquin@workstation: ~/ansible-lab
iquin@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playbook.yaml

PLAY [First Ansible Playbook] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]
ok: [centos1]

TASK [Ping all managed nodes] *****
ok: [server1]
ok: [server2]
ok: [centos1]

TASK [Display Hello World] *****
ok: [server1] => {
  "msg": "Hello World from server1"
}
ok: [server2] => {
  "msg": "Hello World from server2"
}
ok: [centos1] => {
  "msg": "Hello World from centos1"
}

TASK [Display Operating System] *****
ok: [server1] => {
  "msg": "Operating System: Ubuntu"
}
ok: [server2] => {
```

```

iquin@workstation: ~/ansible-lab
ok: [server1] => {
  "msg": "Hello World from server1"
}
ok: [server2] => {
  "msg": "Hello World from server2"
}
ok: [centos1] => {
  "msg": "Hello World from centos1"
}

TASK [Display Operating System] *****
ok: [server1] => {
  "msg": "Operating System: Ubuntu"
}
ok: [server2] => {
  "msg": "Operating System: Ubuntu"
}
ok: [centos1] => {
  "msg": "Operating System: CentOS"
}

PLAY RECAP *****
centos1                : ok=4    changed=0    unreachable=0    failed=0    skipped=0
  rescued=0    ignored=0
server1                : ok=4    changed=0    unreachable=0    failed=0    skipped=0
  rescued=0    ignored=0
server2                : ok=4    changed=0    unreachable=0    failed=0    skipped=0
  rescued=0    ignored=0
iquin@workstation:~/ansible-lab$

```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

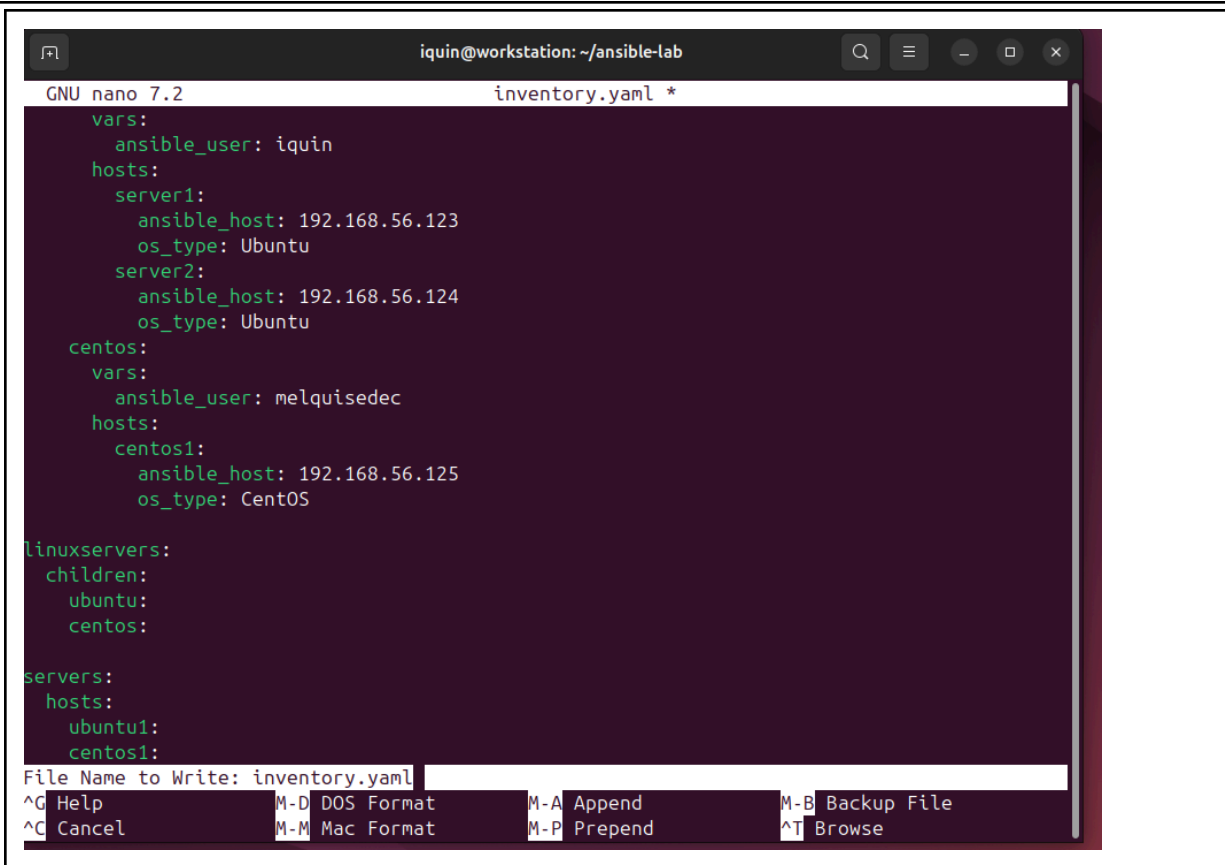
Create a new group named servers that contains only:

- ubuntu1
- centos1

Create a playbook named challenge.yaml that displays the message:

Welcome to Ansible Laboratory

Run the playbook and demonstrate that only hosts in the servers group receive the message



```
GNU nano 7.2                               inventory.yaml *
vars:
  ansible_user: iquin
hosts:
  server1:
    ansible_host: 192.168.56.123
    os_type: Ubuntu
  server2:
    ansible_host: 192.168.56.124
    os_type: Ubuntu
centos:
  vars:
    ansible_user: melquisedec
  hosts:
    centos1:
      ansible_host: 192.168.56.125
      os_type: CentOS

linuxservers:
  children:
    ubuntu:
    centos:

servers:
  hosts:
    ubuntu1:
    centos1:
```

File Name to Write: inventory.yaml

^G Help	M-D DOS Format	M-A Append	M-B Backup File
^C Cancel	M-M Mac Format	M-P Prepend	^T Browse

```
iquin@workstation: ~/ansible-lab
centos1 : ok=2 changed=0 unreachable=0 failed=0 skipped=0
rescued=0 ignored=0
ubuntu1 : ok=0 changed=0 unreachable=1 failed=0 skipped=0
rescued=0 ignored=0

iquin@workstation:~/ansible-lab$ nano inventory.yaml
iquin@workstation:~/ansible-lab$
iquin@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.yaml

PLAY [Display variables] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [centos1]

TASK [Ansible Laboratory] *****
ok: [server1] => {
  "msg": "Welcome to Ansible Laboratory"
}
ok: [centos1] => {
  "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP *****
centos1 : ok=2 changed=0 unreachable=0 failed=0 skipped=0
rescued=0 ignored=0
server1 : ok=2 changed=0 unreachable=0 failed=0 skipped=0
rescued=0 ignored=0

iquin@workstation:~/ansible-lab$
```

Guide Questions

1. What is the role of the Control Node in Ansible?
 - The role of the control node in ansible is basically the head or core that controls everything and keeps everything in check, somewhat like a head of the whole system.
2. Why is an inventory file required?
 - Well an inventory file is required mainly because its how they get reached, like the IP and usernames in the host files. Without it ansible wouldnt know which machine to target when running the commands.
3. What is the purpose of a metagroup?
 - This is a group that contains other groups that let you manage them together, it saves us time from repeating the same task separately
4. How do variables simplify Ansible administration?
 - It lets you store reusable values like usernames or file paths instead of hardcoding it into every task. This makes the playbook more flexible and you can change variables instead of editing multiple lines.
5. What is the difference between an ad hoc command and a playbook?
 - Its an one off ansible command that run directly from the terminal for quick tasks, like checking a connectivity with a ping. Where a playbook is a written file containing multiple ordered tasks that can be reused and version controlled.

Conclusion:

- As for my conclusion for this activity, i have learned so many things about ansible and how each of

them work together, especially the yaml files which helps the system find where they're located and how they work together, and honestly it was very hard and stressful due to multiple errors and indentation markers which makes the whole system corrupts and not work. But in the end I managed to fix and debug it.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.